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CUET UG

Biology

Microbes in Human Welfare

Official Syllabus Unit VIII · Class 12 · CUET UG 2027 Cycle Notes

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Syllabus Alignment & Exam Blueprint

Sub-Topic	Official NTA CUET UG Syllabus Scope	High-Yield Exam Focus
Household Food Processing	Microbes in household products: Curd, dough, bread, traditional foods, cheese varieties.	LAB (Vitamin B_{12} boost); *S. cerevisiae* (CO_2 puffing); Swiss cheese (*P. sharmanii*) vs Roquefort (*P. roqueforti*).
Industrial Fermentation	Microbes in industrial products: Fermented beverages, distillation, commercial organic acids, and enzymes.	Brewer's yeast; distilled (whisky/rum) vs non-distilled (wine/beer); *A. niger* (citric), *A. aceti* (acetic); lipases & pectinases.
Bioactive Molecules & Drugs	Antibiotics (penicillin discovery) and bioactive molecules (streptokinase, cyclosporin A, statins).	Streptokinase (clot buster); Cyclosporin A (*T. polysporum*, immunosuppressant); Statins (*M. purpureus*, HMG-CoA reductase).
Sewage Treatment (STP)	Microbes in sewage treatment: Primary (physical) vs Secondary (biological) treatment; BOD dynamics.	Sequential filtration & grit settling; Aeration tank & **Flocs**; **BOD** metric; Activated sludge & river action plans (GAP/YAP).
Biogas Generation	Microbes in production of biogas; methanogenic bacteria, cattle rumen, and biogas plant design.	*Methanobacterium* (anaerobic); Biogas mix ($CH_4 + CO_2 + H_2S$); Rumen cellulose breakdown; IARI & KVIC plant.
Biocontrol Agents	Microbes as biocontrol agents: Natural predators, Bt endotoxin, and baculoviruses in IPM.	Ladybird (aphids); Dragonfly (mosquitoes); *Bacillus thuringiensis* (Bt cry toxin); Baculoviruses (NPV narrow-spectrum).
Biofertilisers	Microbes as biofertilisers: Symbiotic/free-living nitrogen fixers, mycorrhiza, and cyanobacteria.	*Rhizobium* vs *Azotobacter*/*Azospirillum*; Mycorrhiza (*Glomus* phosphorus uptake); Cyanobacteria (*Anabaena*, *Nostoc*) in paddy.

CUET HIGH-YIELD EXAM ARCHITECTURE & WEIGHTAGE GUIDE

Microbes in Human Welfare is one of the most reliable and high-scoring memory-based chapters in CUET UG Biology, yielding **3 to 5 compulsory questions (12–20 marks)** per test session. Questions focus strictly on precise two-part matching pairs (Organism → Product/Function): (1) Household fermentation and cheese makers (*P. sharmanii* vs *P. roqueforti*); (2) Organic acids and their source fungal/bacterial species; (3) The "Holy Trinity" of bioactive molecules: Streptokinase (clot buster), Cyclosporin A (immunosuppressant), and Statins (HMG-CoA reductase inhibitor); (4) Sewage treatment sequence, flocs composition, and BOD relationships; and (5) Specific ecological attributes of Baculoviruses (NPV) and Mycorrhiza (*Glomus*).

1. Microbes in Household Food Processing

Microorganisms—encompassing bacteria, fungi, protozoa, and microscopic viruses—are ubiquitous across Earth's biosphere, inhabiting extreme environments ranging from deep geothermal vents ($> 100^{\circ}$ C) to Antarctic permafrost and acidic soils. While public perception frequently associates microbes exclusively with infectious pathogens, the overwhelming majority of microbial taxa perform indispensable beneficial roles in human nutrition, industrial manufacturing, wastewater recycling, ecological nutrient cycling, and agriculture.

Lactic Acid Bacteria (LAB) and Curd Production

- **Bacterial Strains:** The biological conversion of fresh milk into curd is orchestrated by a group of beneficial bacteria collectively termed **Lactic Acid Bacteria (LAB)**, predominantly including ***Lactobacillus acidophilus***, *Lactobacillus bulgaricus*, and *Streptococcus thermophilus*.
- **The Inoculum / Starter:** A small quantity of fresh curd is added to lukewarm milk (35° C $- 40^{\circ}$ C) as an **inoculum** or **starter**, which introduces millions of active LAB cells into the substrate.
- **Biochemical Transformation:**
  1. LAB metabolize the disaccharide milk sugar (**lactose**) into **lactic acid** through homofermentative or heterofermentative pathways:
  $$C_{12}H_{22}O_{11} + H_2O \rightarrow 4C_3H_6O_3 \quad \text{(Lactic Acid)}$$
  2. The production of lactic acid decreases the pH of milk (pH $\approx 6.6 \rightarrow 4.5$).
  3. The lowered pH coagulates and **partially digests the principal milk protein, casein**, converting liquid milk into a semi-solid, gelatinous curd matrix.
- **Nutritional & Clinical Significance:**
  - **Vitamin B_{12} Enrichment:** Fermentation by LAB significantly enhances the nutritional density of milk by synthesizing substantial quantities of **Vitamin B_{12}** (cyanocobalamin).
  - **Gut Protective Barrier:** In the human stomach and intestinal tract, colonized LAB secrete antimicrobial bacteriocins and maintain an acidic microenvironment that actively inhibits the proliferation of harmful pathogenic microbes.

Fermented Foods: Dosa, Idli, Bread & Toddy

- **Dosa and Idli Dough Leavening:**
  - The semi-liquid batter used to prepare traditional South Indian delicacies such as dosa and idli is prepared by soaking and grinding rice and black gram (*urad dal*).
  - The batter is naturally fermented by resident airborne and seed-borne bacteria, primarily ***Leuconostoc mesenteroides*** and ***Streptococcus faecalis***.
  - **Puffed-Up Spongy Texture:** The batter rises and develops a characteristic puffed-up appearance due to the extensive production and entrapment of **carbon dioxide gas (CO_2)** generated during microbial anaerobic respiration.
- **Bread Making (Baker's Yeast):**
  - The leavening of wheat flour dough in commercial bread making is driven by the single-celled budding yeast ***Saccharomyces cerevisiae***, universally designated as **Baker's Yeast**.
  - Baker's yeast ferments fermentable maltose and sucrose into ethyl alcohol and carbon dioxide:
  $$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$$
  - The released CO_2 gas expands within the viscoelastic gluten protein network of wheat flour. When baked at high oven temperatures (200° C), the gas pockets expand further before setting, and the alcohol evaporates, leaving light, porous, and spongy bread.
- **Traditional Beverages: Toddy:**
  - **Toddy** is a traditional refreshing alcoholic beverage consumed widely across coastal South India.
  - It is prepared by collecting the clear sugary sap exudate from tapped unopened spadices of palm trees (*Caryota urens* or *Borassus flabellifer*), which is naturally fermented by wild yeasts and LAB over several hours into a mildly alcoholic drink.

Traditional & Industrial Cheeses

Cheese is one of the oldest microbial food preservation techniques known to human civilization. Different commercial cheese varieties are characterized by distinct textures, aromas, hole sizes, and flavors dictated by the specific ripening microbial culture employed:

1. **Swiss Cheese:** Characterized by large, conspicuous holes ("eyes"). These dramatic cavities are carved out by massive volumes of **carbon dioxide (CO_2)** gas synthesized during the slow secondary fermentation of lactic acid by the bacterium ***Propionibacterium freudenreichii* subsp. *shermanii***.
2. **Roquefort Cheese:** A semi-hard, intensely flavored blue-veined cheese crafted from sheep's milk. It is ripened by inoculating specific spores of the filamentous mold ***Penicillium roqueforti***, whose lipolytic and proteolytic enzymes permeate the curd to impart its unique tangy flavor and bluish-green marbling.
3. **Camembert Cheese:** A creamy, soft-ripened surface-mold cheese ripened by the fungal mycelium of ***Penicillium camemberti***.

2. Industrial Fermentation & Fermented Beverages

When microorganisms are exploited to manufacture valuable biochemicals, therapeutic proteins, or beverages on an industrial scale, microbial cultivation must be scaled up from laboratory glassware to enormous industrial culture vessels termed **Fermentors** or **Bioreactors**, which hold capacities ranging from hundreds to tens of thousands of liters.

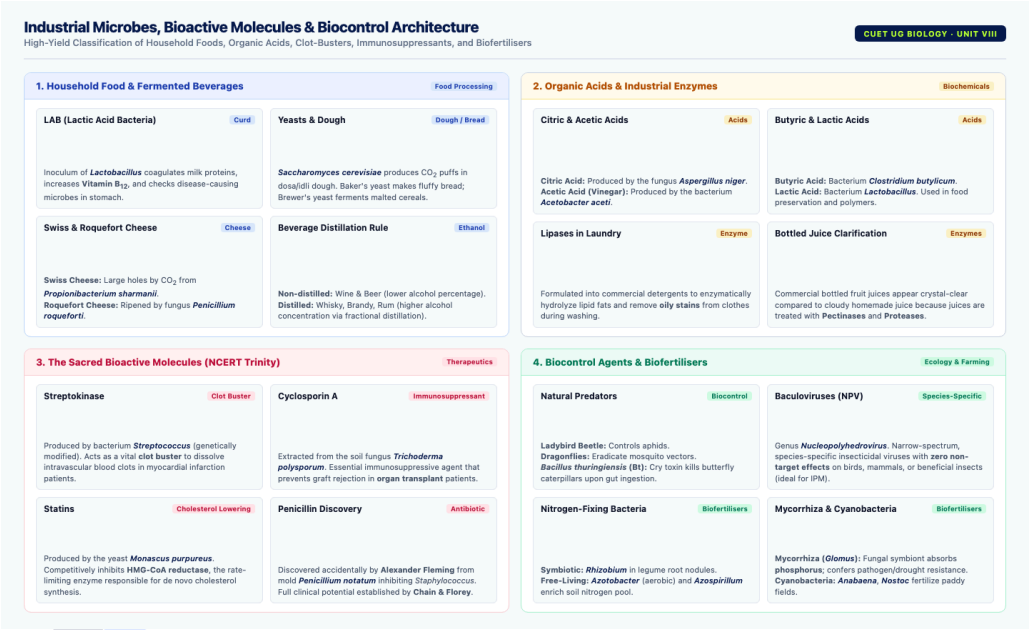


Figure 1: Industrial microbiology & agro-ecological biotechnology — microbial fermentations for organic acids, enzymes, and bioactive molecules alongside biofertiliser nitrogen-fixing networks.

Alcoholic Fermented Beverages

- **Brewer's Yeast:** Industrial ethanolic fermentation utilizes selected high-yielding strains of ***Saccharomyces cerevisiae***, historically celebrated as **Brewer's Yeast**.
- **Substrates:** Fermentation employs diverse carbohydrate-rich feedstocks, including malted barley, wheat, rye, maize, crushed grapes, and molasses.
- **Distilled vs Non-Distilled Beverages (Top NTA Exam Distinction):**
  - The maximum alcohol concentration achievable through direct yeast fermentation alone is strictly limited ($\approx 13\% - 15\%$) because rising ethanol concentrations become toxic and autolyze the yeast cells.
  - **Non-Distilled Beverages:** Produced by direct fermentation of fruit juice or malted grain wash **without any subsequent distillation**. Consequently, they maintain lower alcohol percentages:
    - **Wine** (fermented from crushed grapes, $\approx 9\% - 12\%$ alcohol).
    - **Beer** (fermented from malted barley grains, $\approx 3\% - 6\%$ alcohol).
  - **Distilled Beverages (Spirits):** The crude fermented broth is subjected to fractional **distillation**, boiling off and concentrating the volatile ethanol vapors to achieve substantially higher proof and alcohol concentrations ($40\% - 50\%$):
    - **Whisky** (distilled from fermented cereal mash).
    - **Brandy** (distilled from fermented grape wine).
    - **Rum** (distilled from fermented sugarcane molasses).

3. Antibiotics: Discovery & Clinical Triumph

Antibiotics—derived from the Greek terms *anti* ("against") and *bios* ("life")—are chemical substances synthesized by certain microorganisms that, in trace concentrations, kill or inhibit the growth of pathogenic microorganisms. In relation to human pathogens, antibiotics represent pro-life therapeutics.

Discovery of Penicillin: Fleming's Serendipity (1928)

- **The Serendipitous Observation:** Sir Alexander Fleming, working at St. Mary's Hospital in London in 1928, was conducting laboratory research on the pathogenic bacterium ***Staphylococcus aureus***.
- Upon inspecting culture plates that had been left unwashed on his laboratory bench, Fleming observed a contaminated mold growing on one agar dish.
- Significantly, around the growing mold colony, there was a distinct, clear zone where **staphylococcal bacterial colonies had completely lysed and failed to grow**.
- Fleming deduced that the mold secreted a diffusible chemical substance that inhibited bacterial cell wall synthesis. He named this active compound **Penicillin**, isolated from the mold ***Penicillium notatum***.
- **Full Therapeutic Potential & The 1945 Nobel Prize:**
  - Fleming published his findings in 1929 but was unable to purify or stabilize the chemical molecule.
  - More than a decade later, during World War II, **Sir Howard Florey** and **Ernst Boris Chain** successfully extracted, concentrated, and established penicillin as an extraordinarily powerful, non-toxic clinical antibiotic.
  - Penicillin was administered on a mass scale to treat Allied soldiers wounded in the European and Pacific theaters of World War II, preventing fatal gangrene and septicemia.
  - In **1945**, the Nobel Prize in Physiology or Medicine was jointly conferred upon **Alexander Fleming, Howard Florey, and Ernst Chain**.
- **Clinical Triumph Over Historical Scourges:** Antibiotics transformed global public health, eradicating millions of premature deaths previously inflicted by bacterial scourges such as **Plague** (*Yersinia pestis*), **Whooping Cough / Kali Khansi** (*Bordetella pertussis*), **Diphtheria / Gal Ghotu** (*Corynebacterium diphtheriae*), and **Leprosy / Kusht Rog** (*Mycobacterium leprae*).

4. Organic Acids, Industrial Enzymes & Bioactive Molecules

Modern industrial biotechnology harnesses specific bacterial and fungal strains as microscopic chemical factories to synthesize concentrated organic acids, biocatalysts, and life-saving pharmaceutical bioactive molecules:

Industrial Organic Acids

Organic Acid	Microbial Source	Taxonomic Class & Industrial Application
Citric Acid	***Aspergillus niger***	Filamentous Ascomycete fungus; food acidulant, preservative, and buffer.
Acetic Acid	***Acetobacter aceti***	Obligate aerobic bacterium; commercial vinegar and chemical solvent.
Butyric Acid	***Clostridium butylicum***	Spore-forming obligate anaerobic bacterium; ester synthesis and flavoring.
Lactic Acid	***Lactobacillus*** sp.	Gram-positive bacterium; food preservation, pharmaceuticals, and bioplastics.

Industrial Enzymes

1. **Lipases:** Synthesized commercially from microbial cultures (such as *Candida* and *Pseudomonas*). Lipases catalyze the hydrolysis of ester bonds in triglycerides. They are incorporated into modern laundry **detergent formulations** to enzymatically degrade and remove stubborn hydrophobic oil and grease stains from fabrics.
2. **Pectinases and Proteases:**
   - Freshly squeezed homemade fruit juices appear cloudy and murky due to suspended insoluble cell-wall pectin fragments and colloidal plant proteins.
   - Industrial juice processors treat raw pressed juices with **pectinases** (from *Aspergillus niger*) and **proteases**.

- These enzymes hydrolyze the structural pectin and protein polymers into soluble micromolecules, producing sparkling, crystal-clear bottled fruit juices.

3. **Streptokinase ("Clot Buster"):**
   - Synthesized by the bacterium ***Streptococcus*** and subsequently modified by recombinant DNA technology.
   - In clinical medicine, streptokinase acts as a potent fibrinolytic agent: it converts systemic plasminogen into active plasmin, which enzymatic breaks down intravascular fibrin clots (thrombi).
   - Administered as an emergency intravenous **"clot buster"** to dissolve occluding thrombi in patients experiencing acute myocardial infarction (heart attack).

Life-Saving Bioactive Molecules

1. **Cyclosporin A (Immunosuppressive Miracle):**
   - A lipophilic cyclic non-ribosomal peptide containing 11 amino acids, isolated from the filamentous soil fungus ***Trichoderma polysporum***.
   - **Clinical Mode of Action:** Cyclosporin A binds to the intracellular receptor cyclophilin, forming a drug-receptor complex that selectively inhibits calcineurin phosphatase. This suppresses the transcription of interleukin-2 in helper T-lymphocytes, preventing immune cell proliferation.
   - **Medical Indication:** Indispensable **immunosuppressive drug** administered to patients undergoing solid organ transplantation (kidney, liver, heart, lung) to prevent graft rejection.
2. **Statins (Blood Cholesterol-Lowering Agents):**
   - Synthesized by the red mold yeast ***Monascus purpureus***.
   - **Mode of Action:** Statins are structural analogues of HMG-CoA that act as **competitive inhibitors** of the rate-limiting hepatic enzyme **HMG-CoA reductase** (β-hydroxy-β-methylglutaryl-CoA reductase), blocking endogenous cholesterol biosynthesis.
   - Extensively prescribed globally (e.g., lovastatin, atorvastatin) to lower serum LDL cholesterol and avert coronary atherosclerosis.

5. Microbes in Sewage Treatment (STP): Primary & Secondary Treatment

Municipal wastewater—commonly termed **sewage**—is generated daily in millions of liters across urban centers. Sewage contains massive quantities of human excreta, household greywater, dissolved organic pollutants, and pathogenic enteric bacteria. Raw municipal sewage cannot be discharged directly into natural watercourses like rivers or lakes because high organic loads rapidly deplete dissolved oxygen, leading to catastrophic aquatic mortality. Sewage must undergo sequential treatment in **Sewage Treatment Plants (STPs)**, primarily utilizing heterotrophic microbes.

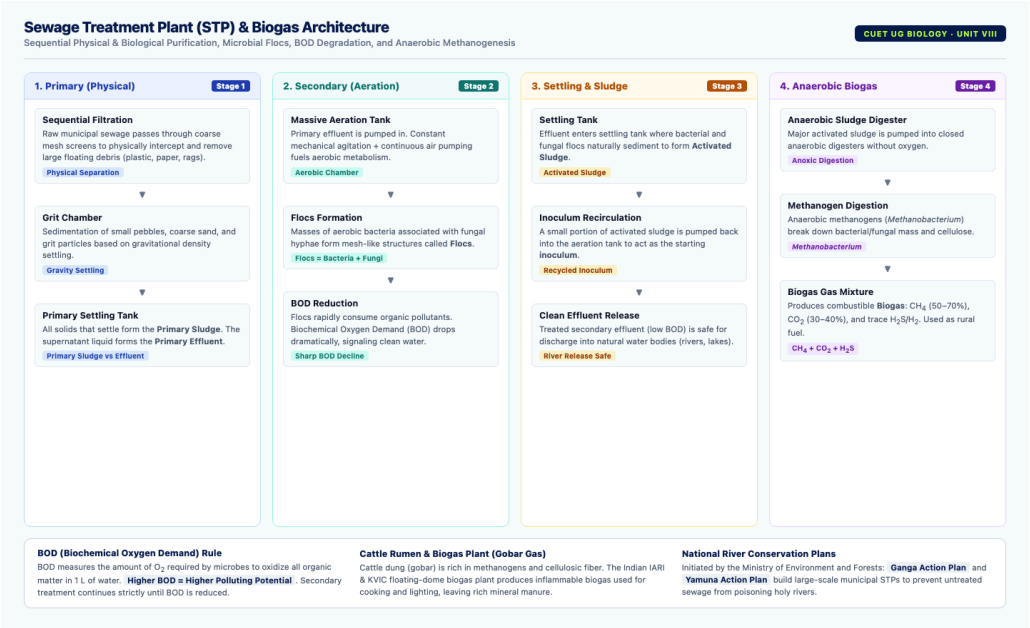


Figure 2: Sewage treatment architecture & biogas kinetics — sequential mechanical primary treatment, aerobic secondary aeration with microbial flocs, and anaerobic sludge digestion generating methane biogas.

Primary Treatment (Physical Separation Phase)

Primary treatment involves the physical removal of large and small solid particles from sewage through sequential physical operations:

1. **Sequential Filtration:** Raw influent sewage is passed through a succession of wire mesh screens to intercept and filter out floating physical debris (plastics, rags, paper, twigs, and leaves).
2. **Grit Removal & Primary Sedimentation:**
   - The screened liquid flows into a **grit settling chamber**, where water velocity is reduced to permit heavy inorganic grit (sand, pebbles, gravel, and silt) to settle gravitationally.
   - The sewage then enters large **primary settling tanks**, where settleable organic solids sink to form the **primary sludge**.
   - The floating oily scum is skimmed off, and the clarified supernatant constitutes the **primary effluent**, which proceeds to biological treatment.

Secondary Treatment (Aerobic Biological Treatment & Flocs)

Secondary treatment is an entirely biological process that utilizes aerobic microbial ecology to consume dissolved organic matter:

- **Aeration Tanks:** The primary effluent is channeled into massive, open **aeration tanks**, where it is continuously mechanically agitated by submerged aerators and supplied with sterile compressed air.
- **Formation of Flocs:** Continuous agitation and vigorous aeration foster the rapid, dense growth of aerobic heterotrophic microorganisms into **flocs**.
  - **NCERT Definition of Flocs: Masses of bacteria intimately associated with fungal filaments to form mesh-like or net-like microscopic structures**.
- **Reduction of Biochemical Oxygen Demand (BOD):**
  - As the flocs rapidly proliferate, they consume the bulk of the dissolved organic matter in the effluent as carbon and energy sources for cellular respiration.
  - This microbial consumption dramatically reduces the **Biochemical Oxygen Demand (BOD)** of the effluent.

The Biochemical Oxygen Demand (BOD) Metric

- **Scientific Definition: Biochemical Oxygen Demand (BOD)** is defined as the total quantity of dissolved oxygen (in milligrams per liter) that would be consumed if all the biodegradable organic matter present in one liter of water were completely oxidized by aerobic microorganisms.
- **The Direct Relationship:** The BOD test measures the rate of oxygen uptake by microbial populations in a water sample over a standardized 5-day incubation at 20° C (BOD_5).

- **The Diagnostic Pollution Metric: There is a direct, linear relationship between BOD and organic pollution: a high BOD indicates heavy organic contamination and severe polluting potential, whereas a low BOD signifies clean, unpolluted water**.
- Secondary treatment is continued vigorously until the BOD of the sewage is drastically reduced by 80% – 90%.

6. Secondary Settling, Anaerobic Sludge Digestion & Biogas

Once the BOD of the aeration tank effluent has been satisfactorily reduced, the treated liquor undergoes final clarification and sludge digestion:

The Settling Tank & Activated Sludge

- The effluent containing microbial flocs is pumped into a quiescent **settling tank**.
- In this quiescent tank, the bacterial-fungal flocs settle rapidly to the bottom by gravitational sedimentation. This sedimented biological mass is termed **activated sludge**.
- **Dual Fate of Activated Sludge:**
  1. **Inoculum Recycle:** A small portion ($\approx 5\% - 10\%$) of the activated sludge is pumped back upstream into the aeration tank to serve as an active **inoculum** (starter), replenishing the aerobic microbial population.
  2. **Anaerobic Sludge Digestion:** The major remaining portion of the activated sludge is pumped into massive, sealed, airtight concrete tanks termed **anaerobic sludge digesters**.

Anaerobic Sludge Digesters & Biogas Generation

- Inside the anaerobic digesters, an entirely different suite of microorganisms—**obligate and facultative anaerobic bacteria**—takes over.
- These anaerobic bacteria hydrolyze and digest the aerobic bacteria and fungi composing the activated sludge.
- **Biogas Composition:** During this anaerobic biochemical digestion, methanogenic bacteria produce a mixture of gases collectively termed **Biogas**:
  - **Methane (CH_4):** $50\% - 70\%$ (the primary combustible, energy-rich component).
  - **Carbon Dioxide (CO_2):** $30\% - 40\%$.
  - Trace amounts of **Hydrogen Sulfide (H_2S)**, Nitrogen (N_2), and Hydrogen (H_2).
- **Dual Utility:** Biogas is inflammable and is captured as a source of clean, renewable thermal and electrical energy for the sewage facility. The treated secondary effluent is disinfected (via chlorination, UV irradiation, or ozonation) and safely discharged into natural rivers.

National River Action Plans: GAP and YAP

With accelerating urban industrialization, the volume of municipal sewage generated daily vastly outpaced the treatment capacity of existing STPs across India, leading to severe ecological degradation of major national waterways. To restore the water quality of holy rivers, the Ministry of Environment, Forest and Climate Change launched comprehensive river action plans:

- **Ganga Action Plan (GAP):** Launched in **1985** by Prime Minister Rajiv Gandhi.
- **Yamuna Action Plan (YAP):** Initiated subsequently to intercept, divert, and treat municipal sewage from major cities located along the Yamuna river basin before industrial effluent enters the river.

7. Microbes in Biogas Generation & Rural Energy Systems

Biogas is a combustible hydrocarbon gas mixture produced by the anaerobic microbial breakdown of biological organic matter (animal dung, agricultural crop residues, kitchen waste, and sewage).

Methanogenic Bacteria & Methanogenesis

- While aerobic oxidation of carbohydrates yields CO_2 and H_2O, certain specialized anaerobic bacteria growing on cellulosic substrates produce substantial volumes of **methane (CH_4)** along with CO_2 and H_2.
- These bacteria are designated as **Methanogens**, and their premier representative is ***Methanobacterium***.
- **Ecological Niches of Methanogens:**

1. Deep anaerobic sludge layers in municipal sewage treatment digesters.
2. Flooded marshlands and waterlogged paddy fields (responsible for marsh gas emissions).
3. **The Rumen of Ruminant Cattle:** A significant fraction of ruminant livestock diet consists of cellulosic plant forage. Methanogens naturally inhabit the **rumen** of cattle, where they synthesize cellulolytic enzymes that break down cellulose, playing an indispensable symbiotic role in bovine nutrition.

- **Gobar Gas:** Consequently, bovine excreta—commonly called cattle dung or **gobar**—is extraordinarily rich in living methanogenic bacteria. In rural India, biogas generated from cattle dung is traditionally termed **Gobar Gas**.

Structural Architecture of a Rural Biogas Plant

A typical household or community biogas plant (standardized by Indian research institutions) consists of:

1. **The Digester Pit:** A circular, 10–15 feet deep brick or concrete well sealed to ensure complete anoxic conditions.
2. **Inlet & Mixing Chamber:** Cattle dung and water are blended in an equal ratio (1:1) in an above-ground mixing tank to produce a viscous **dung slurry**, which flows down through an inclined inlet pipe into the bottom of the digester.
3. **Floating Gas Holder:** A cylindrical steel or fiberglass bell placed inverted over the digester slurry. As methanogens metabolize and generate gas, the holder rises upward, providing a visual gauge of stored gas pressure.
4. **Gas Outlet Pipe:** A control valve and pipe mounted on the gas holder deliver biogas directly to rural households for cooking and illumination.
5. **Sludge Outlet & Fertilizer Reservoir:** The spent, digested slurry is displaced upward and discharges through an overflow outlet. This spent slurry is rich in nitrogen, phosphorus, and potassium, serving as premium organic manure for farming.

- **Pioneering Indian Institutions:** The technology of rural biogas production was developed and disseminated in India primarily through the collaborative efforts of the **Indian Agricultural Research Institute (IARI)** and the **Khadi and Village Industries Commission (KVIC)**.

8. Microbes as Biocontrol Agents: Sustainable Agro-Ecology

Modern industrial agriculture has relied excessively upon synthetic chemical pesticides, insecticides, and weedicides. Chemical pesticides are indiscriminate toxins that kill beneficial pollinators (bees), predatory insects, and earthworms alongside targeted pests, bioaccumulate through trophic levels (biomagnification), pollute groundwater tables, and leave toxic residues on food crops. **Biological control (Biocontrol)** is an agro-ecological strategy that manages agricultural pests and plant diseases using natural biological predation and parasitism rather than chemical warfare.

The Holistic Philosophy of Organic Farming

Organic farming does not seek the complete, sterile eradication of pests. Instead, it maintains pests at manageable, sub-economic threshold levels through a vibrant, biodiverse ecological web of predatory checks and balances. Total eradication of pests is ecologically disastrous because beneficial predators and parasitoids starve and disappear, leaving the field vulnerable to catastrophic secondary pest outbreaks.

Natural Predators and Microbial Biocontrol Agents

Biocontrol Agent	Biological Class	Target Pest Managed & Ecological Mechanism
Ladybird Beetle	Predatory Coleopteran insect	Controls soft-bodied **aphids** infesting agricultural crops through voracious predation.
Dragonflies	Predatory Odonate insect	Controls aquatic larvae and flying adults of disease-carrying **mosquitoes**.
Bacillus thuringiensis (Bt)	Spore-forming soil bacterium	Bt spores mixed with water and sprayed onto brassicas; ingested by **butterfly caterpillars**, endotoxins dissolve in alkaline gut, killing larvae.
Trichoderma sp.	Free-living soil Ascomycete fungus	Colonizes rhizosphere root ecosystems; secretes chitinases to suppress **soil-borne**

Biocontrol Agent	Biological Class	Target Pest Managed & Ecological Mechanism
		fungal root pathogens (e.g., *Fusarium*, *Rhizoctonia*).
Baculoviruses (NPV)	Double-stranded DNA viral pathogens	Belong to genus ***Nucleopolyhedrovirus***; highly effective **species-specific, narrow-spectrum insecticidal agents** targeting agricultural arthropods.

Baculoviruses (*Nucleopolyhedrovirus* / NPV)

- **Host Specificity:** Baculoviruses are insect-pathogenic viruses belonging primarily to the genus ***Nucleopolyhedrovirus***.
- **The Precision Paradigm:** They are celebrated in modern Integrated Pest Management (IPM) because they exhibit strict **species-specific, narrow-spectrum insecticidal activity**.
- **Zero Non-Target Toxicity:** They have been experimentally verified to cause **no negative impacts whatsoever on plants, mammals, birds, fish, or non-target beneficial insects** (such as honeybees and ladybirds).
- **Conservation of Ecosystems:** NPVs are especially desirable in ecologically sensitive zones and organic agricultural frameworks where beneficial parasitoids must be preserved while suppressing specific lepidopteran caterpillars.

9. Microbes as Biofertilisers: Biological Nutrient Fixation

Indiscriminate reliance on synthetic chemical fertilizers (urea, diammonium phosphate) has depleted soil organic humus, altered soil pH, destroyed earthworm populations, and triggered lethal eutrophication of freshwater lakes. **Biofertilisers** are selected living microorganisms that enrich the nutrient quality and structural fertility of agricultural soils through biological nitrogen fixation, phosphorus solubilization, and organic humus mineralization.

The Three Primary Taxonomic Groups of Biofertilisers

1. **Bacteria (Nitrogen-Fixing Systems):**
   - **Symbiotic Bacteria:** The classic paradigm is ***Rhizobium***, which establishes root nodule symbioses with leguminous plants (pulses, beans, clover). The enzyme nitrogenase inside nodule bacteroids reduces atmospheric nitrogen into bioavailable ammonium compounds. Another vital symbiotic actinomycete bacterium is ***Frankia***, which forms nitrogen-fixing root nodules on the roots of non-leguminous trees (such as *Alnus* and *Casuarina*).
   - **Free-Living (Non-Symbiotic) Soil Bacteria:** Aerobic bacteria such as ***Azotobacter*** and ***Beijerinckia***, as well as anaerobic bacteria such as ***Clostridium***, fix atmospheric nitrogen independently within soil pores, enriching soil nitrogen without host plants.
   - **Associative Symbionts:** Bacteria such as ***Azospirillum*** live in close rhizosphere association with the roots of cereal grasses (maize, sorghum, sugarcane), substantially enhancing grain yields.
2. **Fungi (Mycorrhizal Mutualisms):**
   - **Mycorrhiza** is a mutualistic symbiotic association established between fungal hyphae and the roots of higher seed plants.
   - Many mycorrhizal species belong to the widespread fungal genus ***Glomus***.
   - **Biological Benefits to Host Plants:**
     - The extensive, microscopic fungal hyphal network acts as a physical extension of the root surface, absorbing **phosphorus** from inaccessible soil mineral complexes and translocating it directly into host root vascular tissues.
     - Imparts heightened **resistance to root-borne fungal and bacterial pathogens**.
     - Confers heightened **tolerance to soil salinity and extreme drought**.
     - Enhances overall plant vigor, vegetative growth, flowering, and reproductive biomass.
     - In exchange, the plant provides the heterotrophic fungus with carbon assimilates (sugars) and sheltered cortical root niches.
3. **Cyanobacteria (Blue-Green Algae):**

- Cyanobacteria are autotrophic, photosynthetic prokaryotes widespread in both aquatic and terrestrial ecosystems, capable of harvesting solar energy while fixing atmospheric nitrogen in specialized, thick-walled cells termed **heterocysts**.
- Key nitrogen-fixing genera include ***Anabaena***, ***Nostoc***, and ***Oscillatoria***.
- **The Paddy (Rice) Field Biofertiliser:** In flooded paddy agricultural ecosystems, cyanobacteria serve as the principal natural biofertiliser. A celebrated mutualism occurs between ***Anabaena azollae*** and the tiny floating freshwater aquatic fern ***Azolla pinnata***. When *Azolla* is grown as green manure in rice paddies, it multiplies rapidly, fixing substantial quantities of nitrogen and contributing vital organic matter, significantly reducing chemical nitrogen requirements.

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Strategic Synthesis: Microbes in Human Welfare

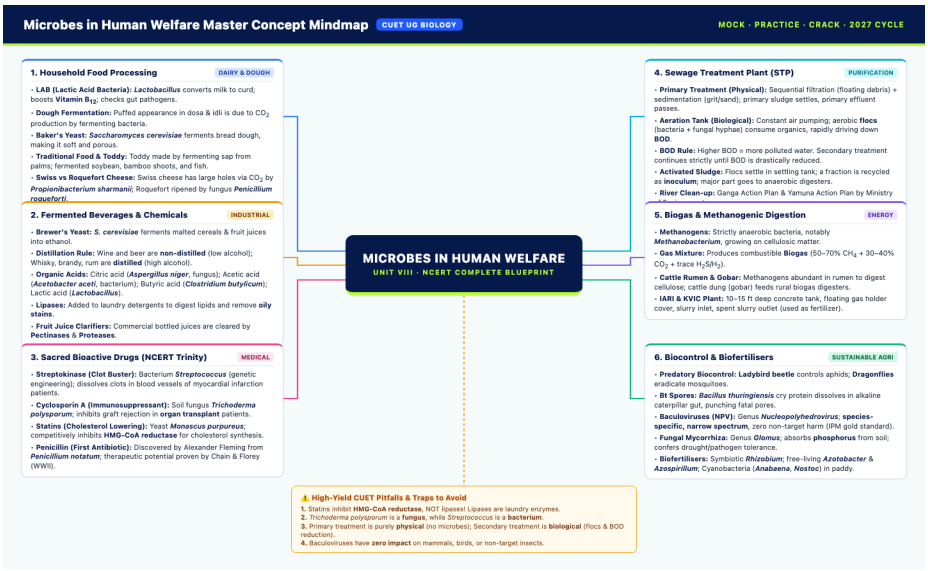


Figure 3: Master synthesis mindmap — high-yield integrated architecture of Microbes in Human Welfare covering household fermentations, industrial chemicals, antibiotics, STP sewage processing, biogas, biocontrol, and biofertilisers.

CUET HIGH-YIELD MEMORY HOOKS & DIAGNOSTIC TRIAD

1. **Household & Industrial Fermentation Triad:** LAB coagulates milk casein and enriches Vitamin B_{12} *Saccharomyces cerevisiae* drives bread leavening (CO_2) and alcoholic fermentation; wine and beer are un-distilled, whereas whisky, brandy, and rum are distilled; Swiss cheese holes are created by CO_2 from *Propionibacterium freudenreichii* subsp. *shermanii*; Roquefort cheese is ripened by *Penicillium roqueforti*.
2. **Biochemicals & Biomedical Therapeutics Triad:** Citric acid (*Aspergillus niger*), Acetic acid (*Acetobacter aceti*), Butyric acid (*Clostridium butylicum*), Lactic acid (*Lactobacillus*); Streptokinase from *Streptococcus* acts as an emergency myocardial "clot buster"; Cyclosporin A from *Trichoderma polysporum* is an immunosuppressive drug; Statins from *Monascus purpureus* competitively inhibit HMG-CoA reductase to lower blood cholesterol.
3. **Ecological Sewage, Biogas & Biofertiliser Triad:** Aerobic aeration flocs (bacteria + fungi) consume organic waste to drastically lower BOD; anaerobic digestion of activated sludge yields Biogas ($CH_4 + CO_2$); *Trichoderma* protects roots from fungi; Baculoviruses (*Nucleopolyhedrovirus*) offer species-specific, narrow-spectrum insecticidal control with zero non-target toxicity; Mycorrhizae (*Glomus*) absorb phosphorus; Cyanobacteria (*Anabaena*, *Nostoc*) fertilize paddy fields.

Master Diagnostic Cheatsheet & Fatal CUET Traps

MASTER CONCEPTUAL & DIAGNOSTIC CHEATSHEET — MICROBES IN HUMAN WELFARE

- **Lactic Acid Bacteria (LAB):** Ferment lactose to lactic acid, coagulating milk casein into curd; significantly increases **Vitamin B_{12}**; checks harmful stomach microbes.
- **Fermented Food Stuffs:** Dosa/idli batter rises due to CO_2 by *Leuconostoc*; bread dough leavened by Baker's yeast (***Saccharomyces cerevisiae***); Swiss cheese large holes carved by CO_2 from ***Propionibacterium freudenreichii* subsp. *shermanii***; Roquefort cheese ripened by the mold ***Penicillium roqueforti***.
- **Alcoholic Distillation:** Wine and Beer are **un-distilled** (lower alcohol); Whisky, Brandy, and Rum are concentrated via **distillation** of fermented wash.
- **Penicillin Discovery:** Sir Alexander Fleming (1928) isolated penicillin from mold *Penicillium notatum* inhibiting *Staphylococcus*; therapeutic potential proven by Howard Florey and Ernst Chain (treating WWII soldiers); 1945 Nobel Prize to Fleming, Florey, and Chain.
- **Industrial Chemicals:** Citric acid (***Aspergillus niger*** - fungus); Acetic acid (***Acetobacter aceti*** - bacterium); Butyric acid (***Clostridium butylicum*** - bacterium); Lactic acid (***Lactobacillus*** - bacterium).
- **Industrial Enzymes & Bioactive Drugs:** Lipases in laundry detergents; Pectinases & Proteases for clarifying bottled fruit juices; **Streptokinase** (*Streptococcus*) as emergency myocardial "clot buster"; **Cyclosporin A** (*Trichoderma polysporum*) as organ transplant immunosuppressant; **Statins** (*Monascus purpureus*) as blood cholesterol-lowering competitive inhibitors of HMG-CoA reductase.
- **Sewage Treatment (STP):** Primary treatment removes physical solids via sequential filtration and grit sedimentation; Secondary treatment uses aerobic **flocs** (bacteria + fungi) to drastically reduce **Biochemical Oxygen Demand (BOD)**; high BOD = heavy organic pollution.
- **Sludge Digestion & Biogas:** Settled flocs form **activated sludge**; digested by anaerobic methanogens (***Methanobacterium***) to produce **Biogas** (50–70% CH_4, 30–40% CO_2, trace H_2S); rural biogas technology developed by IARI and KVIC.
- **Biocontrol Agents:** Ladybird beetles kill aphids; dragonflies kill mosquitoes; Bt spores kill butterfly caterpillars; *Trichoderma* suppresses root fungal pathogens; **Baculoviruses (NPV)** provide species-specific, narrow-spectrum insecticidal control with zero non-target toxicity (ideal for IPM).
- **Biofertilisers:** Nitrogen fixers: symbiotic (***Rhizobium***, ***Frankia*** in non-legume *Alnus*), free-living (***Azotobacter***, ***Beijerinckia***, ***Clostridium***), associative (***Azospirillum***); Mycorrhiza (***Glomus***) absorbs **phosphorus**; Cyanobacteria (***Anabaena***, ***Nostoc***, ***Oscillatoria***) fertilize paddy fields.

THE SEVEN FATAL CUET TRAPS IN MICROBES IN HUMAN WELFARE

1. **The Distilled vs Non-Distilled Trap:** Never confuse beverages! Wine and Beer are produced **without distillation**; Whisky, Brandy, and Rum are produced **with distillation**.
2. **The Chemical Source Taxonomy Trap:** *Aspergillus niger* is a **fungus** (producing citric acid); *Acetobacter aceti* and *Clostridium butylicum* are **bacteria**; *Monascus purpureus* is a **yeast/fungus** (producing statins).
3. **The Swiss Cheese Gas Trap:** The large holes in Swiss cheese are caused by CO_2 gas released by the bacterium *Propionibacterium freudenreichii* subsp. *shermanii*, NOT by yeast or fungi!
4. **The BOD Pollution Metric Trap:** Remember: **Higher BOD indicates greater organic pollution** and higher polluting potential. Low BOD indicates clean water.
5. **The Flocs Composition Trap:** Flocs consist of **aerobic bacteria associated with fungal filaments**, NOT anaerobic microbes.
6. **The Baculovirus Target Trap:** Baculoviruses of the genus *Nucleopolyhedrovirus* have **zero negative impacts** on plants, mammals, birds, fish, or non-target insects. They are strictly species-specific and narrow-spectrum.
7. **The Mycorrhizal Mineral Trap:** Mycorrhizal fungi (genus *Glomus*) specifically absorb **phosphorus** from the soil and pass it to host plants, NOT nitrogen!

Practice Questions — Part 1 (Q1 to Q5)

Q1. During the conversion of milk into curd, Lactic Acid Bacteria (LAB) induce biochemical transformations that enhance both curd texture and human health. Which statement correctly identifies a primary nutritional benefit imparted by LAB?

(A) They significantly enhance the nutritional value of milk by synthesizing substantial amounts of Vitamin B_{12}
(B) They ferment lactose into ascorbic acid, significantly increasing the Vitamin C content of the curd
(C) They degrade toxic casein proteins into synthetic glycogen granules stored in the intestinal mucosa
(D) They eliminate all resident gastric flora by secreting high concentrations of concentrated hydrochloric acid

Sol. Q1. When milk is converted into curd by Lactic Acid Bacteria (LAB), predominantly *Lactobacillus*, the bacteria ferment lactose into lactic acid, which coagulates and partially digests casein milk proteins. In addition to creating curd, this fermentation significantly enhances the nutritional value of the milk by increasing its **Vitamin B_{12}** content. LAB also play a protective role in the human stomach by checking pathogenic microbes. **Answer: (A)**

Q2. The puffed-up, spongy appearance observed during the leavening of bread dough and South Indian idli/dosa batter is primarily caused by:

(A) The evaporation of ethanol vapors during room temperature dough storage
(B) The generation of carbon dioxide (CO_2) gas by anaerobic microbial respiration trapped in the dough
(C) The mechanical churning of gluten proteins during high-speed industrial kneading
(D) The secretion of proteolytic enzymes that digest starch into insoluble nitrogen gas

Sol. Q2. Bread dough rises and develops a soft, spongy texture due to alcoholic fermentation mediated by Baker's yeast, ***Saccharomyces cerevisiae***. The yeast metabolizes sugars to produce ethanol and **carbon dioxide (CO_2)** gas. As the CO_2 bubbles expand and become trapped within the gluten network, they cause the characteristic puffed-up appearance of the dough. **Answer: (B)**

Q3. Which of the following organic acids is correctly matched with its industrial microbial source organism?

(A) Citric acid — *Acetobacter aceti* (Bacterium)
(B) Acetic acid — *Clostridium butylicum* (Fungus)
(C) Citric acid — *Aspergillus niger* (Fungus)
(D) Butyric acid — *Lactobacillus* (Yeast)

Sol. Q3. The precise verified NCERT pairings are: (1) **Citric acid** is produced by the filamentous fungus ***Aspergillus niger***; (2) **Acetic acid** is produced by the bacterium ***Acetobacter aceti***; (3) **Butyric acid** is produced by the obligate anaerobic bacterium ***Clostridium butylicum***; and (4) **Lactic acid** is produced by the bacterium ***Lactobacillus***. Therefore, option C is correctly matched. **Answer: (C)**

Q4. Statins, a premier class of pharmacological agents prescribed globally to lower serum cholesterol, function biochemically by which mechanism?

(A) Competitively inhibiting the enzyme HMG-CoA reductase responsible for cholesterol synthesis
(B) Directly hydrolyzing dietary cholesterol into bile pigments in the duodenal lumen
(C) Irreversibly binding to low-density lipoproteins and precipitating them into the gallbladder
(D) Allosterically activating pancreatic lipases to accelerate the excretion of fatty acids

Sol. Q4. Statins are bioactive molecules produced by the red mold yeast ***Monascus purpureus***. Statins act as **competitive inhibitors** of the enzyme **HMG-CoA reductase**, which catalyzes the rate-limiting step in the hepatic biosynthesis of cholesterol. By occupying the active site of HMG-CoA reductase, statins prevent the conversion of HMG-CoA to mevalonate, effectively lowering blood cholesterol levels. **Answer: (A)**

Q5. Biochemical Oxygen Demand (BOD) is a standard regulatory parameter evaluated in wastewater analytics. Which statement regarding BOD is scientifically accurate?

(A) A high BOD value signifies that water is completely purified and safe for drinking
(B) BOD evaluates the quantity of inorganic heavy metals settled gravitationally in the grit chamber
(C) BOD is completely independent of the quantity of dissolved organic matter present in sewage
(D) A high BOD value directly indicates a high concentration of organic matter and high polluting potential

Sol. Q5. Biochemical Oxygen Demand (BOD) refers to the amount of dissolved oxygen required by aerobic microbes to oxidize all biodegradable organic matter in one liter of water. Because microbes consume oxygen in direct proportion to the quantity of organic substrate present, a **high BOD directly signifies heavy organic pollution and high polluting potential**. Once secondary treatment successfully reduces the organic load, the BOD drops significantly. **Answer: (D)**

Practice Questions — Part 2 (Q6 to Q10)

Q6. During the secondary (biological) stage of sewage treatment in an aeration tank, what biological structure is formed to consume dissolved organic matter?

(A) Flocs, composed of masses of aerobic bacteria associated with fungal filaments to form mesh-like networks
(B) Primary sludge, composed of gravitational sediments of pebbles and floating plastics
(C) Anaerobic biofilms, composed of pure cultures of methanogenic archaebacteria
(D) Mycorrhizal root nodules, which absorb dissolved nitrogenous phosphates directly from effluent

Sol. Q6. In secondary sewage treatment aeration tanks, continuous air supply and agitation foster the growth of **flocs**, which are strictly defined in NCERT as **masses of bacteria associated with fungal filaments to form mesh-like structures**. As these aerobic flocs metabolize, they consume the dissolved organic matter in the effluent, dramatically lowering its Biochemical Oxygen Demand (BOD). **Answer: (A)**

Q7. Biogas is a combustible fuel produced in anaerobic sludge digesters and rural gobar gas plants. Which microbial group and primary gas constituent correctly describe biogas?

(A) Aerobic nitrifying bacteria producing sulfur dioxide (SO_2) as the major combustible component
(B) Anaerobic methanogenic bacteria producing methane (CH_4) as the major combustible component
(C) Photosynthetic cyanobacteria producing ozone (O_3) as the major combustible fuel component
(D) Facultative heterotrophic yeasts producing pure carbon monoxide (CO) as the primary constituent

Sol. Q7. Methanogens, such as ***Methanobacterium***, are obligate anaerobes that grow **strictly in the absence of molecular oxygen**. In anaerobic sludge digesters and cattle dung plants, they digest cellulosic organic materials to produce **Biogas**, which contains $50\% - 70\%$ **methane (CH_4)** as its primary combustible constituent, along with $30\% - 40\% CO_2$ and trace quantities of H_2S and H_2. **Answer: (B)**

Q8. A farmer wishes to protect crop roots against soil-borne fungal pathogens without applying synthetic chemical fungicides. Which biocontrol fungus should be introduced into the root rhizosphere?

(A) *Penicillium notatum*
(B) *Aspergillus niger*
(C) *Trichoderma* species
(D) *Monascus purpureus*

Sol. Q8. ***Trichoderma*** is a genus of free-living, soil-dwelling filamentous fungi that thrive in the root ecosystems (rhizosphere). They are highly effective **biocontrol agents** against several fungal plant root pathogens, suppressing disease through competitive rhizosphere colonization, mycoparasitism, and the secretion of antimicrobial hydrolytic enzymes. **Answer: (C)**

Q9. Root nodules of leguminous plants host symbiotic bacteria capable of converting atmospheric nitrogen into bioavailable ammonia. This vital biofertiliser bacterium is:

(A) *Azotobacter*
(B) *Azospirillum*
(C) *Oscillatoria*
(D) *Rhizobium*

Sol. Q9. The Gram-negative bacterium ***Rhizobium*** establishes a specific symbiotic association with the roots of leguminous plants (pulses, beans, peas), developing specialized root nodules. Inside these anaerobic nodules, the bacterial enzyme nitrogenase reduces atmospheric dinitrogen (N_2) into ammonia, supplying vital organic nitrogen to the host plant while receiving carbohydrates. **Answer: (D)**

Q10. A patient is rushed to the hospital emergency department exhibiting acute myocardial infarction caused by an occluding coronary thrombus. The emergency physician should immediately administer which recombinant enzyme as a "clot buster"?

(A) Lipase
(B) Streptokinase
(C) Pectinase
(D) Protease

Sol. Q10. **Streptokinase** is an enzyme produced by the bacterium ***Streptococcus*** and subsequently modified by genetic engineering. It functions as a potent **clot buster**, activating plasminogen to dissolve intravascular fibrin clots (thrombi) that occlude coronary blood vessels, thereby restoring myocardial blood perfusion in acute heart attack patients. **Answer: (B)**

Practice Questions — Part 3 (Q11 to Q15)

Q11. The large, conspicuous holes observed in traditional Swiss cheese are carved out by which microorganism producing copious amounts of carbon dioxide (CO_2)?

(A) *Propionibacterium freudenreichii* subsp. *shermanii* (B) *Penicillium roqueforti*
(C) *Saccharomyces cerevisiae* (D) *Streptococcus thermophilus*

Sol. Q11. The large holes characteristic of Swiss cheese are produced by the bacterium ***Propionibacterium freudenreichii* subsp. *shermanii***. During the slow ripening process, this bacterium ferments lactic acid to produce propionic acid and large volumes of **carbon dioxide (CO_2)** gas. As bubbles of CO_2 accumulate and escape through the firm cheese matrix, they carve out the conspicuous holes. **Answer: (A)**

Q12. Commercially bottled fruit juices purchased from retail stores appear sparkling and crystal-clear compared to cloudy homemade fruit juices because industrial juices are treated with:

(A) Lipases and Amylases (B) Cellulases and Ligases
(C) Pectinases and Proteases (D) Streptokinase and Lysozyme

Sol. Q12. Fresh homemade fruit juices appear cloudy and turbid due to suspended insoluble plant cell-wall pectins and proteinaceous colloids. Industrial fruit juice manufacturers treat the raw pressed juice with the hydrolytic enzymes **Pectinases** and **Proteases**. These enzymes hydrolyze the structural pectin and protein polymers into soluble fragments, rendering bottled juices crystal-clear. **Answer: (C)**

Q13. Cyclosporin A is a cyclic non-ribosomal peptide widely administered in organ transplantation as an immunosuppressive agent. It is extracted commercially from which organism?

(A) *Monascus purpureus* (Yeast) (B) *Streptococcus* (Bacterium)
(C) *Trichoderma polysporum* (Fungus) (D) *Aspergillus niger* (Fungus)

Sol. Q13. **Cyclosporin A** is an 11-amino-acid cyclic peptide isolated from the soil fungus ***Trichoderma polysporum***. It acts as a powerful **immunosuppressive agent** by inhibiting calcineurin phosphatase and suppressing T-lymphocyte activation. It is administered to patients undergoing organ transplantation (e.g., kidney, liver, heart) to prevent host immune rejection of the grafted organ. **Answer: (C)**

Q14. Baculoviruses belonging to the genus *Nucleopolyhedrovirus* (NPV) are considered premier biocontrol agents in Integrated Pest Management (IPM) because they:

(A) Produce broad-spectrum toxins that kill all insects, fish, and birds in the agricultural zone
(B) Are obligate plant parasites that stimulate root development and mineral phosphorus uptake
(C) Infect human pathogens in domestic sewage to reduce biochemical oxygen demand
(D) Are strictly species-specific and narrow-spectrum, exhibiting zero toxicity to mammals, birds, fish, or non-target insects

Sol. Q14. Baculoviruses of the genus ***Nucleopolyhedrovirus*** (NPV) are celebrated in sustainable agro-ecology because they are strictly **species-specific and narrow-spectrum insecticidal pathogens**. They infect specific arthropods without causing any harm to plants, mammals, birds, fish, or non-target insects such as bees and predatory beetles, making them ideal for Integrated Pest Management (IPM). **Answer: (D)**

Q15. In terrestrial ecosystems, many plant families establish a mutualistic mycorrhizal association with the fungal genus *Glomus*. What critical nutritional benefit does the fungal partner provide to the host plant?

(A) Direct fixation of atmospheric elemental dinitrogen into ammonium ions
(B) Absorption of phosphorus from the soil and its translocation into the host plant
(C) Synthesis of broad-spectrum antibiotics that permanently sterilize the leaf phyllosphere
(D) Excretion of massive quantities of ethyl alcohol that protects roots from freezing

Sol. Q15. In a mycorrhizal mutualism, such as those formed by the fungal genus ***Glomus***, the extensive fungal mycelial network acts as an extension of the plant's root system, absorbing **phosphorus** from the soil and translocating it directly to the host plant. Additionally, mycorrhizal associations confer heightened resistance to root-borne pathogens and enhance tolerance to drought and salinity. **Answer: (B)**

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